



VIT[®]

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MINI PROJECT

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Simulation and Statistical Verification of Bivariate Gaussian Random Vectors using Cholesky Decomposition (Stock Price & Market Index Model)

ABSTRACT

This mini project models the relationship between a stock price and a market index using a bivariate Gaussian distribution. In financial systems, stock prices often move in correlation with market indices such as NIFTY or SENSEX. This project simulates such correlated behavior using Cholesky decomposition. The joint probability density function (PDF) is computed and visualized using 3D surface and contour plots. The empirical mean and covariance obtained from simulated data are compared with theoretical values to validate the model.

INTRODUCTION

In financial markets, stock prices are not independent; they are influenced by broader market movements. For example, when the market index rises, many stocks also tend to rise. This dependency can be modeled using a bivariate Gaussian distribution.

In this project:

- X_1 represents the **market index (e.g., NIFTY)**
- X_2 represents a **stock price (e.g., a company stock)**

We simulate their joint behavior using statistical techniques and verify the correctness of the model.

OBJECTIVES

The main objectives of this project are:

1. To model stock price and market index as correlated Gaussian variables.
2. To generate 5000 samples using Cholesky decomposition.
3. To compute the joint probability density function.
4. To visualize the distribution using 3D and contour plots.
5. To verify statistical accuracy using mean and covariance comparison.

THEORETICAL BACKGROUND

Financial Interpretation of Variables

$X_1 \rightarrow$ Market Index (e.g., NIFTY), $X_2 \rightarrow$ Stock Price

Mean Vector: $\mu = [1$
2]

Interpretation:

- Mean of market index = 1 (normalized value)
- Mean of stock price = 2 (normalized value)

Covariance Matrix: $C = [1 \ 0.8$
0.8 2]

Interpretation:

- Market index variance = 1
- Stock price variance = 2
- Covariance = 0.8 $\rightarrow$ Strong positive correlation

This means:

When the market index increases, the stock price also tends to increase.

Joint Probability Density Function

$$f(x) = 1 / (2\pi\sqrt{|C|}) * \exp(-1/2 (x - \mu)^T C^{-1} (x - \mu))$$

This function represents the probability distribution of stock price and market index together.

Cholesky Decomposition

To simulate correlated financial data:

$$C = LL^T \quad X = LZ + \mu$$

Where:

- Z = independent random shocks (market noise)
- L = correlation structure
- X = simulated stock and index values

METHODOLOGY

The following procedure was used in this project:

Step 1: Define mean and covariance (financial model)

Step 2: Apply Cholesky decomposition

Step 3: Generate random market shocks

Step 4: Introduce correlation

Step 5: Generate stock & index data

Step 6: Compute joint PDF

Step 7: Plot 3D surface

Step 8: Plot contour diagram

Step 9: Compute empirical statistics

Step 10: Compare with theoretical values.

MATLAB IMPLEMENTATION

Code

```
%% Bivariate Gaussian Simulation - Stock Price & Market Index Model
```

```
clear; clc; close all;
```

```
%% 1. Define Theoretical Parameters
```

```
mu = [1; 2];           % Mean: [Market Index, Stock Price]
```

```
C = [1, 0.8; 0.8, 2];   % Covariance Matrix
```

```
N = 5000;               % Number of samples
```

```
%% 2. Generate Correlated Samples (Cholesky Method)
```

```
L = chol(C, 'lower');    % Cholesky decomposition
```

```
Z = randn(2, N);         % Independent standard normal variables
```

```
X = L * Z + mu;          % Correlated samples
```

```
X_samples = X';          % Convert to Nx2 format
```

```
%% 3. Manual PDF Calculation (Theoretical Joint PDF)
```

```
x1 = linspace(mu(1)-4, mu(1)+4, 100);
```

```
x2 = linspace(mu(2)-4, mu(2)+4, 100);
```

```
[X1, X2] = meshgrid(x1, x2);
```

```
% Pre-allocate PDF matrix
```

```
Z_pdf = zeros(size(X1));
```

```
% Determinant and inverse of covariance matrix
```

```
detC = det(C);
```

```
invC = inv(C);
```

```
% Loop to calculate PDF at each grid point manually
```

```
for i = 1:size(X1, 1)
```

```
    for j = 1:size(X1, 2)
```

```
        x_vec = [X1(i,j); X2(i,j)] - mu;
```

```
        exponent = -0.5 * (x_vec' * invC * x_vec);
```

```

        normalization = 1 / (2 * pi * sqrt(detC));

        Z_pdf(i,j) = normalization * exp(exponent);

    end

end

%% 4. Data Visualization

% Figure 1: 3D Joint PDF Surface

figure('Name', '3D PDF Surface', 'Color', 'w');

surf(X1, X2, Z_pdf);

title('3D Joint PDF of Stock Price and Market Index');

xlabel('Market Index (X_1)');

ylabel('Stock Price (X_2)');

zlabel('Joint PDF f(X_1, X_2)');

shading interp;

colormap(jet);

colorbar;

view(-30, 30);

% Figure 2: Contour Plot and Scatter

figure('Name', 'Contours', 'Color', 'w');

contour(X1, X2, Z_pdf, 20, 'LineWidth', 1.2);

hold on;

scatter(X_samples(1:800, 1), X_samples(1:800, 2), ...

        6, 'k', 'filled', 'MarkerFaceAlpha', 0.2);

title('Contour of Stock Price vs Market Index');

xlabel('Market Index');

ylabel('Stock Price');

grid on;

axis equal;

%% 5. Statistical Results

fprintf('--- Statistical Results ---\n');

```

```

% Empirical Mean

emp_mean = mean(X_samples);

fprintf('Empirical Mean: [%0.4f, %0.4f]\n', emp_mean(1), emp_mean(2));

% Empirical Covariance

emp_cov = cov(X_samples);

disp('Empirical Covariance Matrix:');

disp(emp_cov);

%% 6. Theoretical Results (for comparison)

fprintf('\n--- Theoretical Values ---\n');

fprintf('Theoretical Mean: [%0.4f, %0.4f]\n', mu(1), mu(2));

disp('Theoretical Covariance Matrix:');

disp(C);

```

FLOWCHART

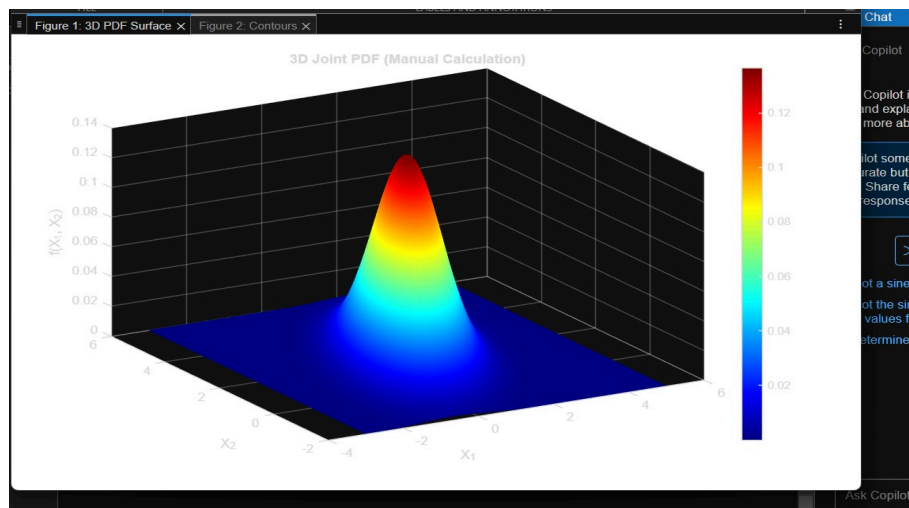
1. Start
2. Define μ and C
3. Perform Cholesky Decomposition
4. Generate Random Samples
5. Generate Correlated Samples
6. Compute Joint PDF
7. Plot 3D Surface
8. Plot Contours
9. Compute Mean and Covariance
10. Compare with Theoretical Values
11. Stop

RESULTS AND DISCUSSION

3D Joint PDF Surface

The 3D plot shows probability density of stock price vs market index.

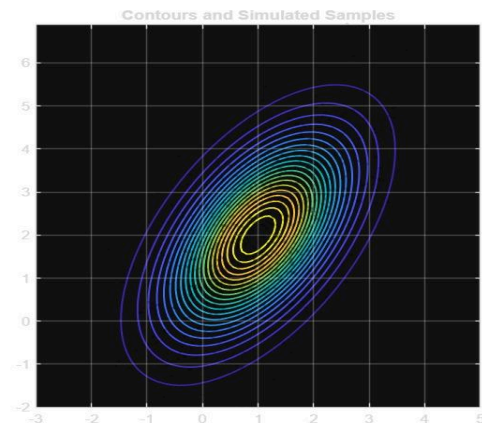
- Peak occurs at mean (1, 2)
- Indicates most probable values



Contour Plot

The contour plot shows elliptical shapes.

- Tilted ellipse $\rightarrow$ positive correlation
- Represents how stock moves with market



Scatter Plot

Simulated points show:

- Upward trend
- Stock price increases with market index

STATISTICAL VERIFICATION

Theoretical Mean

Theoretical Mean = $[1, 2]$

Empirical Mean

Empirical Mean is computed using simulated samples and is approximately equal to theoretical mean. Empirical Mean $\approx [1, 2]$

Theoretical Covariance Matrix

$[1 \ 0.8]$

$[0.8 \ 2]$

Empirical Covariance Matrix

Empirical covariance matrix is approximately equal to theoretical covariance matrix.

This verifies that the generated random vectors follow the desired bivariate Gaussian distribution.

REAL-WORLD INTERPRETATION

This model represents:

- Market index fluctuations (economic conditions)
- Stock price reactions (company performance + market influence)

Positive covariance (0.8) means:

The stock is strongly influenced by the market.

APPLICATIONS

- Stock market prediction
- Portfolio risk analysis
- Financial modeling
- Algorithmic trading
- AI in finance
- Risk management systems

ADVANTAGES OF CHOLESKY METHOD

- Simple financial modeling
- Captures correlation
- Useful for simulations
- Easy to implement in MATLAB

CONCLUSION

This project successfully modeled the relationship between a stock price and a market index using a bivariate Gaussian distribution. Using Cholesky decomposition, correlated data was generated and verified statistically. The results confirmed that the simulated data follows the theoretical distribution. This demonstrates how statistical methods can be applied to financial systems.

SUMMARY

This project simulates and verifies the relationship between a stock price and market index using a bivariate Gaussian model and Cholesky decomposition.

REFERENCES

1. Sheldon Ross – Introduction to Probability and Statistics
2. Miller and Childers – Probability and Random Processes
3. MATLAB Documentation
4. Statistical Signal Processing Lecture Notes

END OF REPORT